

TAYEL AI LABS

Advanced Prompt Engineering

Session 1 — Foundations & First Techniques

Lessons 00 – 04

00 What a model is

01 Anatomy

02 Context

03 Few-shot

04 Reasoning

[View the course on GitHub](#) →

A teaching deck — every slide links to the full lesson.

What we'll cover — and where to read more

Five short lessons take you from "what is this thing" to writing a prompt that works on purpose. No code. Click any lesson to open the full text on GitHub.

00

What a Language Model Actually Is

It predicts text — it does not look things up.

[open](#) →

01

Anatomy of a Prompt

Six components; diagnose a failure by naming the missing one.

[open](#) →

02

Context & Grounding

The model only knows what you put in the prompt.

[open](#) →

03

Examples (Few-Shot)

Showing beats telling; 2-3 examples fix drift.

[open](#) →

04

Reasoning (Chain of Thought)

Let it think in steps before it answers.

[open](#) →

What a Language Model Actually Is

Mental model first. The model continues your text in the most probable way — one token at a time. It does not understand you or look anything up.

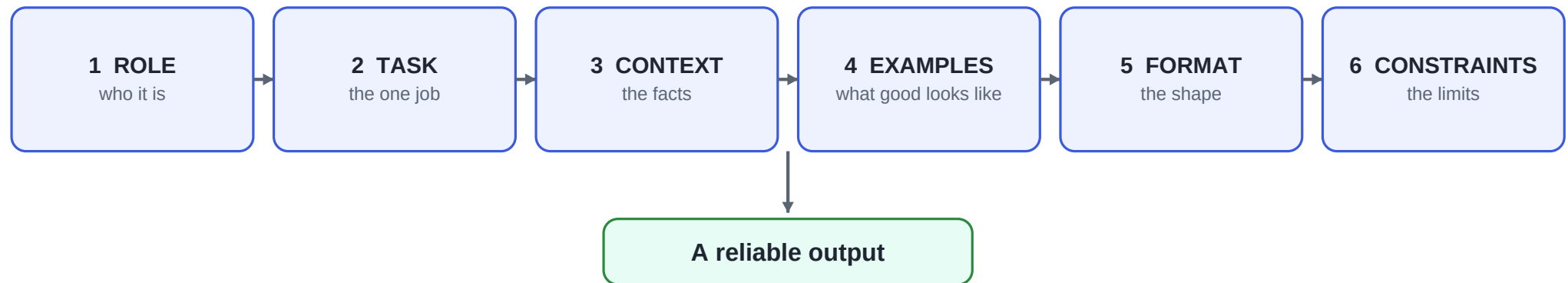
- **Predicts the next token** — "the prompt is the entire world it sees."
- **No memory between chats** — it only knows what's in the prompt right now.
- **Hallucination** — with no fact, it writes a plausible-looking one. Not lying; just text prediction.
- **Tokens** — it reads word-chunks; Arabic costs ~2-3x more tokens than English.

The one habit:

Before sending, ask: "If a smart stranger read ONLY this text, could they produce what I want?" If no, the prompt isn't finished.

Anatomy of a Prompt — the six components

Almost every strong prompt is built from these six ingredients. You rarely need all six — add the ones the task requires.

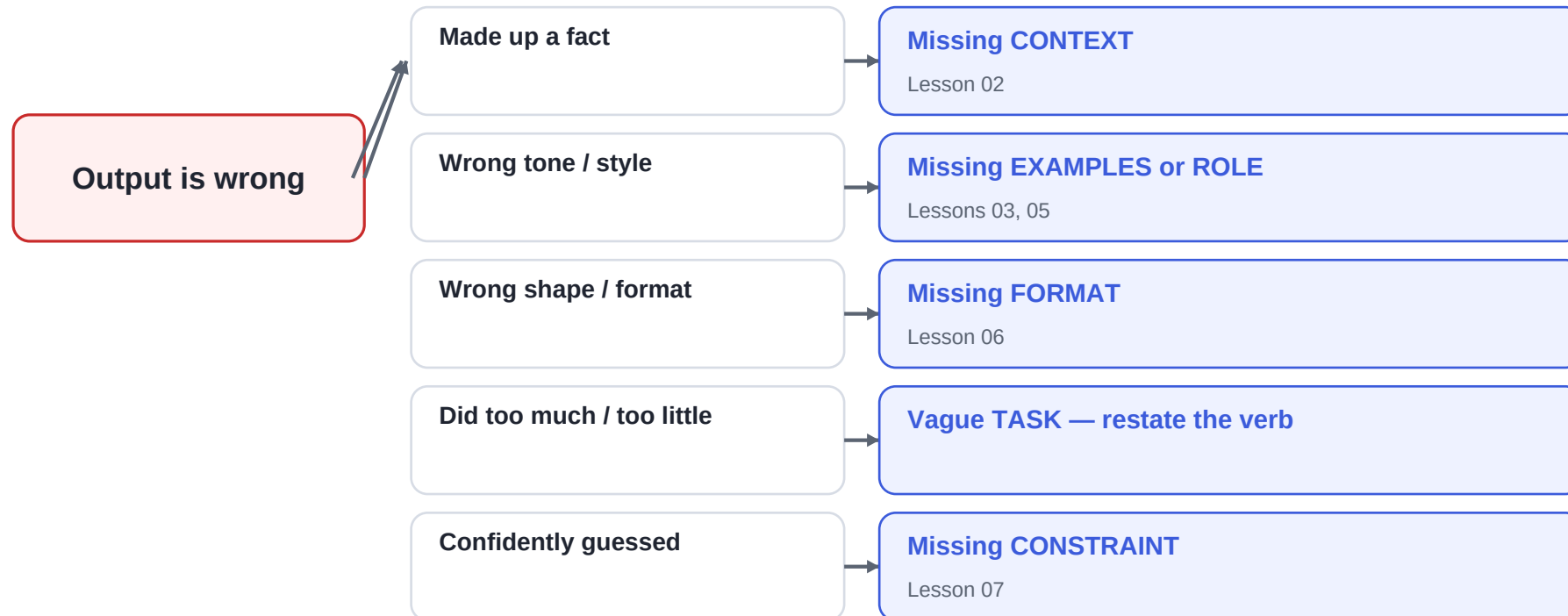


Why this matters:

When output is wrong, don't reword at random. Name the missing ingredient and add just that one. The next slide is the diagnostic map.

Diagnosing a failing prompt

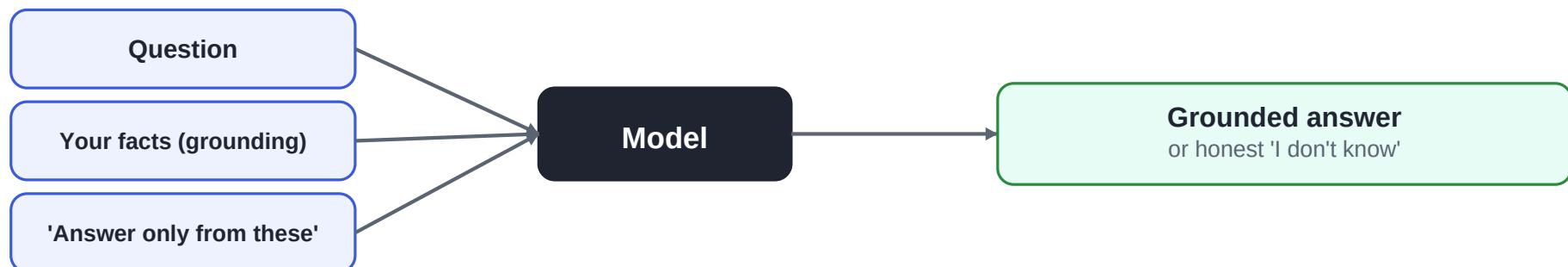
Output is wrong? Ask HOW it's wrong — each symptom points to one missing component.



Context & Grounding

The model knows the AVERAGE, not your specifics. "It made that up" is almost always a missing-context bug. Two moves fix it:

- Put the facts in the prompt, wrapped in delimiters — `<facts> ... </facts>`
- Delimiters separate INSTRUCTIONS (obey) from DATA (only read). Same line as security (Lesson 12).
- Ground it: "Answer ONLY from the facts above. If not covered, say you're unsure — don't guess."



Examples — Few-Shot Prompting

Showing beats telling. Give 2-3 worked examples of input -> desired output and the model continues the pattern — fixing tone and format faster than any description.

ZERO-SHOT



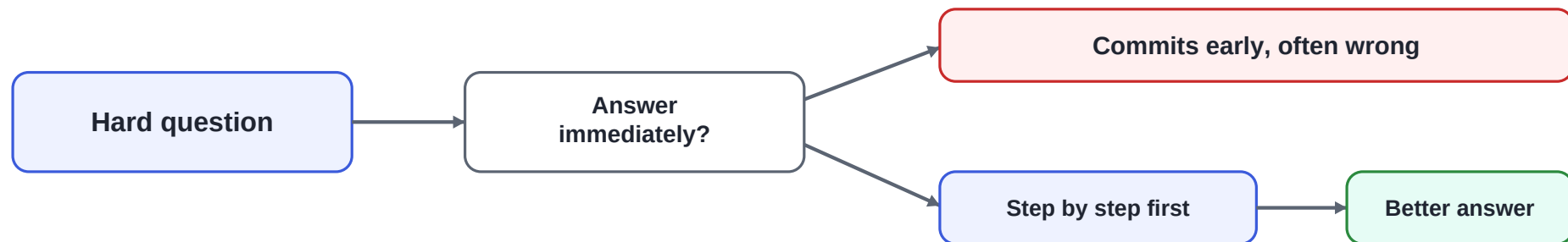
FEW-SHOT



- **Start with two examples; add a third only to cover a case the first two didn't.**
- Put a hard case (sarcasm, Arabic+English mix) IN your examples — the model learns the boundary.
- Your examples ARE the spec — a wrong example is a wrong instruction, followed perfectly.

Reasoning — Chain of Thought

On hard problems (math, multi-condition decisions) let it "think step by step" before answering. Each step becomes context for the next, so it commits less to a wrong first token.



- Reasoning is structured thinking, not just "be detailed" (which only pads).
- Hide the working: ask for a THINKING section, then a REPLY section — show users only REPLY.
- Skip it for simple lookups/transforms — it only adds cost and latency (Lesson 13).

Weak vs Good — the same request, two prompts

WEAK

"Summarise this email."

You still have to read it to know if you must DO anything. Vague world in, vague summary out.

GOOD

You are my inbox assistant. Summarise the email below in one sentence, then list any action it asks of me. If none, say 'No action needed.' Don't invent deadlines. <email>...</email>

role + task + format + context + constraint = a decision, not a chore.

14 more paired examples, one per technique: [examples/weak-vs-good.md](#)

A fill-in-the-blanks scaffold for any new prompt: [templates/prompt-template.md](#)

Assignment — Session 1

Five short parts. No code. You're graded on whether each prompt has the right components and whether your reasoning about WHY is sound.

- **Part 1 — Name the missing components in a weak prompt (Lesson 01).**
- **Part 2 — Rewrite a REAL prompt with all six components + grounding + an escape hatch (01, 02).**
- **Part 3 — Few-shot a style with two examples incl. one edge case (Lesson 03).**
- **Part 4 — Make a task reason step-by-step; compare with/without (Lesson 04).**
- **Part 5 — Diagnose three failures: name the one missing component each (Lesson 01).**

Open the full assignment →

Self-check: run the universal checklist in
[examples/weak-vs-good.md](#)

EVERYTHING, IN ONE PLACE

Links — dive in anytime

The whole course is public on GitHub. Every item below is clickable.

Course home (README + roadmap)

start here



Curriculum — 15 lessons, 4 stages

the full path



3-session teaching plan

how to deliver it



Lesson 00 — What a model is



Lesson 01 — Anatomy of a prompt



Lesson 02 — Context & grounding



Lesson 03 — Few-shot



Lesson 04 — Chain of thought



Weak vs Good examples

the fastest way to learn



Prompt template

copy for any new prompt



Session 1 assignment

hand this in



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